# A Theory of Economic Slack

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# CHAPTER 20.

# Opportunities for future research

This book has presented evidence that slack is ubiquitous in the US economy and developed slackish models of markets and business cycles based on the evidence. There is still a lot that we do not understand about slack, however. In this chapter we map promising avenues for research on slack. We discuss how slack could help us better understand development and business cycles in other countries, and how slack could be measured and modeled better.

## 20.1. Prevalence of slack in other countries

As we saw in chapter 2, the United States might be the economy least hospitable to slack, and yet slack pervades it. It is therefore natural to expect that other countries are at least as slackish as the United States—which implies that the book's theory of slack might be fruitful there too. In this section we review the evidence.

### 20.1.1. Labor market slack in other countries

Unemployment is a universal problem, found on all labor markets (Feldmann 2009, table 1). This is why it is so important to study it, include it into our models, and account for it in policymaking. Unemployment is present throughout the developed world (Elsby, Hobijn, and Sahin 2013). The Beveridge curve is more or less visible in all these countries (Elsby, Michaels, and Ratner 2015). Unemployment is also prevalent in the developing world (O'Higgins 2003, figure 11). Although developing countries do not have more unemployment than developed countries, they experience a much higher share of self-employment (Poschke 2025, figure 1). And self-employment appears to be a mild form of unemployment (Robinson 1936), as workers choose self-employment primarily because they face unemployment in the formal sector (Breza, Kaur, and Shamdasani 2021).

The psychological cost of unemployment has also been confirmed in other developed countries. Just like in the United States, unemployed workers in the United Kingdom suffer from low mental well-being (Clark and Oswald 1994; Blanchflower and Oswald 2004). And they are much more likely to be anxious and depressed and to suffer from low self-esteem—even compared to workers in low-pay jobs (Theodossiou 1998). A typical concern about studies linking unemployment to poor health is that they might not be able to separate between causality (unemployment causes poor health) and selection (people in poor health are more likely to become unemployed). But panel data, following workers over time, are able to identify the causal effect of unemployment on health. For instance, using a large, representative panel dataset of German workers, Winkelmann and Winkelmann (1998) find that unemployment reduces life satisfaction and that the nonpecuniary cost of unemployment is much larger than the pecuniary cost. Using the same panel dataset, Lucas et al. (2004) find that people do not get used to becoming unemployed: workers who have already lost jobs in the past react as badly to a new spell of unemployment as workers who are becoming unemployed for the first time.

Unemployment also seems to have extremely high psychosocial costs in the developing world. A field experiment by Hussam et al. (2022) in Bangladesh illustrates just how large the psychosocial cost of unemployment is. The experiment shows that paid employment raises psychosocial well-being substantially more than the same amount of cash alone. In fact, two-thirds of the workers who were employed in the experiment would be willing to forgo cash payments and continue working for free.

### 20.1.2. Product market slack in other countries

Slack is also present in product markets all over the world, and probably even more in developing countries than in developed countries, so the slackish framework might be especially useful there. Walker et al. (2024, figure 1) provide evidence of slack in firms across the globe, and argue that rates of slack are higher in low-income countries. They also provide specific survey evidence for urban and rural Kenyan firms, which shows widespread product market slack.

A visible source of product market slack is wasted food. UN Environment Programme (2021) documents that food unsold by grocery stores and restaurants generates a significant amount of food waste across the globe. Brancoli (2021) provides a specific example: he explains why bread remains unsold at grocery stores in Sweden and documents how this generates food waste.

Another element suggesting that product market slack is prevalent globally is that sellers strive to form long-term relationships with buyers in all countries. For example, 70% of firms in the Euro area have long-term relationships with their customers (Fabiani et al. 2006, table B3). Firms in developing countries also appear to rely heavily on long-term relationships to operate. Macchiavello (2022, p. 340) reviews a wide range of evidence and concludes that in developing countries:

> Many—and perhaps most—transactions between firms occur in long-term relationships rather than in spot markets, as typically theorized in economic models.

Long-term customer relationships are governed by implicit contracts that alleviate matching problems, as I found in a survey of French bakers in the summer of 2007.[1] First, customer relationships alleviate the uncertainty associated with random demand. A baker told me that demand is difficult to predict and that having a large clientele of loyal customers who make it a habit to purchase bread in the shop was therefore important. In fact, "good" customers are expected to come every day to the bakery. Customer relationships also alleviate the uncertainty associated with random supply. Being a customer means having the assurance that your usual bread will be available, even on days when supply runs low. Of course, this is possible because bakers know exactly what customers order every day through their long association. A baker told me that it would be "unacceptable" to run out of bread for a customer, and that customers would probably "leave the bakery" if that happened.

### 20.1.3. Price and wage rigidities in other countries

The price and wage rigidity used in the book's theory also appear in other countries, which makes a framework with rigid price and wage norms particularly applicable. Indeed, across the globe, prices and wages do not appear to be fully responsive to shocks. Administrative data and household surveys across 16 Western countries show that wages are fairly rigid (Dickens et al. 2007). Wage rigidity is also prevalent in developing countries (Kaur 2019). Large-scale firm surveys show that prices are quite rigid throughout the Euro area (Fabiani et al. 2006). Given such rigidities, we expect fluctuations in slack to be substantial—because adjustments in slack are required to equilibrate supply and demand when prices are rigid.

We argued that price and wage rigidities might be caused by fairness concerns, and we provided evidence from the United States. But there is substantial evidence that people in other countries also care about fairness: Frey and Pommerehne (1993) provides evidence

[1] The survey is described in Eyster, Madarasz, and Michaillat (2021).

from Switzerland and Germany, Gielissen, Dutilh, and Graafland (2008) from the Netherlands, and Boycko and Shiller (2016) from Russia. Firms across the globe also identify fairness as a major concern in price setting. Researchers have surveyed more than 12,000 firms across developed economies about their pricing practices; among the theories that the managers deem most important in explaining price rigidity, some version of "being fair to customers" invariably appears (Eyster, Madarasz, and Michaillat 2021, tables 1 and 2).

More generally, we assumed that social norms determine prices and wages in slackish markets. Social norms determine how people think they should behave and others should behave (Akerlof and Kranton 2010, chapter 2). This means that for instance workers will refuse to work for an employer who violates the norm—who pays a wage that is too low—and workers will be willing to punish other workers who accept to work at a norm-violating wage. Breza, Kaur, and Krishnaswamy (2025) report compelling evidence that all of this occurs in Indian labor markets, even when unemployment is high. Indian workers not only refuse to work at below-normal wages, but they are willing to pay to punish workers who accept such abnormally low wages.

## 20.2. Applying the theory of slack to other countries

In this book we built the theory of slack based on US evidence, calibrated the models to the US economy, and applied the theory to US policies. But, as unemployment and slack exist everywhere, this book's theory should be helpful in other countries too: to understand how markets operate, how business cycles affect the economy, and how policies can be used to improve market outcomes.

### 20.2.1. Labor market policies in other countries

The slackish labor market model developed in this book has already been helpful to understand the effects of various policies in several European countries: for example, in France (Crepon et al. 2013), in Austria (Lalive, Landais, and Zweimuller 2015), and in Sweden (Cheung et al. 2025). It might be helpful to study other policies in comparable labor markets.

The model might also be helpful to understand developing economies and design appropriate policies to combat poverty and foster development, and thus to provide more fulfilling lives to workers everywhere. Indeed, although developing countries see large flows of workers on the labor market (Donovan, Lu, and Schoellman 2023), job rationing in the formal sector is prevalent (Breza, Kaur, and Shamdasani 2021). The model implies for instance that policies that stimulate labor demand, such as public employment, might reduce unemployment effectively. Public employment might be especially potent in slack labor markets, when it crowds out private employment less. The theory provides support for India's public work program, the National Rural Employment Guarantee Act, and suggests that the program might be especially effective in slack local labor markets, just as Breza, Kaur, and Shamdasani (2021, section 8) find.

### 20.2.2. Self-employed economies

In chapter 13, we developed the simplest possible slackish model of an aggregate economy. In that model, all workers are self-employed and sell their production directly to other people. This is not very realistic to describe the modern US economy or other developed economies, where most production is mediated by firms. But it might be useful to describe developing economies because there a large share of the labor force is indeed self employed. As Porta and Shleifer (2014, figure 4) show, the share of workers in the labor force who are self-employed decreases sharply with GDP per capita; the share is close to 100% in the world's poorest countries. The model of chapter 13 might be particularly well adapted to describe the economy of such countries, and to explain why slack is so prevalent there.

### 20.2.3. Business cycle shocks in other countries

In the book we find that fluctuations in unemployment in the United States are driven by labor demand shocks. It would be interesting to assess if this conclusion holds in other countries. Because we reached that conclusion from Okun's law, a first step would be to ascertain whether Okun's law holds in other countries. Ball, Leigh, and Loungani (2017) show that it holds in 20 developed economies, which suggests that labor demand shocks also play a central role there. Ball et al. (2019) extend these results and show that Okun's law holds in many developing economies as well. The slackish business cycle model might therefore be useful in all these economies, as it features labor demand shocks that are nonneutral and produce meaningful unemployment fluctuations that respect Okun's law.

We went one step further and found that these labor demand shocks in fact arise from underlying aggregate demand shocks, not technology shocks. It would be good to know if this conclusion holds in other countries as well. We reached the conclusion by producing Okun plots of short-run output fluctuations against short-run product market slack fluctuations. It would be interesting to produce similar plots in other countries. The challenge will be to obtain good measures of product market slack there.

A related task is to use slack data to obtain better measures of total factor productivity, purified from fluctuations in slack. Although total factor productivity might not be an important driver of business cycles, it is of course the primary driver of economic growth, so measuring it well is important to understand long-run development. As we discussed, measuring total factor productivity accurately requires to measure slack accurately and to properly separate fluctuations in slack from fluctuations in total factor productivity—as both enter fluctuations in measured productivity.

### 20.2.4. FERU in other countries

As the slackish framework developed in the book is likely to be applicable in other countries, the policy tools introduced in the book should also be applicable beyond the United States. Many countries have a full-employment mandate, but no clear full-employment target. Computing the FERU in these countries would be helpful to guide policymakers towards full employment.

Unemployment rate, vacancy rate, and labor market tightness can be computed in other countries in which data on the number of job seekers, number of job vacancies, and number of labor force participants are available. From this the FERU formula $u^* = \sqrt{uv}$ can be applied. And from it it would be possible to compute the unemployment gap $u - u^*$. For instance, Gokten, Heimberger, and Lichtenberger (2024) apply the formula to Germany, Sweden, Austria, Finland, and the United Kingdom, and analyze the evolution of the unemployment gap in these countries since the 1970s. Liargovas et al. (2025) apply the formula and compute the unemployment gap in Greece, 2009–2024. Moving forward, the real-time unemployment gaps could be used to adjust monetary and fiscal policy in response to shocks in these countries, as described in the book.

However, the FERU formula does require assumptions on the cost of unemployment, the cost of recruiting, and the shape of the Beveridge curve. In countries in which these statistics take different values than in the United States, the FERU must be computed from a more general formula. The shape of the Beveridge curve is particularly likely to differ across countries (Gaddnas and Keranen 2023, table 1). In that case, it is necessary to use the generalized formula for the efficient labor market tightness and adjust the calibration of the parameters. Gaddnas and Keranen (2023) take a step in that direction and compute the FERU in Finland, Sweden, Germany, and the Netherlands using country-specific Beveridge elasticities. Germain (2024) compute the FERU in Belgium's three regions—Brussels, Flanders, and Wallonia—using region-specific Beveridge elasticities.

### 20.2.5. Phillips curve in other countries

In the book we build a slackish Phillips curve that links aggregate tightness to inflation. We also show that the Phillips curve ensures that the divine coincidence holds: inflation is on target whenever the economy is at full employment.

There is good evidence in favor of the slackish Phillips curve in the United States, and it would be fascinating to see whether the slackish Phillips curve holds in other countries too: Do we see that slack is a superior indicator of inflationary pressures? And do we see that when the economy is at full employment, inflation is on target? Benigno and Eggertsson (2024) have started to explore these questions. They find similar patterns in other countries, but more research to confirm and expand these findings would be valuable. A key empirical challenge here again is to find measures of slack that are of good quality and that cover a sufficiently long period.

One piece of evidence suggests that the slackish Phillips curve might describe well the behavior of inflation in the developing world. Egger et al. (2022) find that a large fiscal stimulus in Kenya led to a sizeable increase in consumption but a minimal change in price inflation. At the same time, Walker et al. (2024) showed that the Kenyan economy is exceedingly slack. This is consistent with the property of the slackish Phillips curve, which we found to be almost flat in slack economies; hence, even with endogenous inflation, we would expect inflation to increase very little in response to the fiscal stimulus.

### 20.2.6. Detecting recessions in other countries

The Michez+ recession indicator can also be computed for any country in which unemployment and vacancy data are available, but the recession threshold of 0.19pp will need to be recalibrated. In other countries, the recession indicator takes different values, and recessions occur at different times, so the threshold must be recomputed to detect recessions as early and accurately as possible. It might also be possible to optimize how the unemployment and vacancy data are smoothed and combined to best detect recessions in other countries, using the method described in Michaillat (2025). For instance, Sikand and Zhang (2026) show that the approach performs well in Japan.

## 20.3. Collecting better evidence on slack

Because slack is not a core concept in macroeconomic research, statistical agencies have not systematically collected data on it—beyond unemployment. Hence, many measures of slack reported in this book are imperfect in some ways. Rectifying this omission would help improve the theory of slack and the design of slack-based policies. In this section we discuss which type of data and evidence would be especially valuable.

### 20.3.1. Data on labor market slack in the United States

When we computed the full-employment rate of unemployment (FERU), we saw that a key statistic is the amount of man-hours devoted to recruiting by firms. There is little evidence on that, so we used the number of job vacancies reported by firms, together with an estimate of the amount of recruiting per vacancy, to infer the amount of recruiting at any point in time. This lack of evidence is not ideal to measure the FERU and unemployment gap in the past, but it could easily be remedied in the future. The BLS would only need to add one question to the Job Opening and Labor Turnover Survey. The recruiting cost could be measured every month by asking firms to report how many man-hours they devote to recruiting in addition to how many vacancies they are currently advertising. As part of the American Time Use Survey, the BLS already asks unemployed workers how many hours they spend every day searching for a job (Aguiar, Hurst, and Karabarbounis 2013). It should be easy to ask firms the same thing. Moving from job vacancies to recruiting hours would improve the measurement of the unemployment gap and thus the conduct of stabilization policies.

Another key statistic that could influence the FERU is the social cost of unemployment. We have seen substantial evidence from sociology, psychology, public health, and medical science that unemployment is detrimental to physical and especially mental health. But few studies have been able to quantify the social cost of unemployment. One such study, by Borgschulte and Martorell (2018), uses administrative military data to estimate the value of nonwork relative to the value of work. It would be valuable to measure the social value of nonwork for other workers and in other datasets to confirm or refine the findings and ascertain more confidently the social cost of unemployment. Using a wonderful field experiment, Hussam et al. (2022) find dramatic psychosocial costs of unemployment in Bangladesh—to the point where many of the subjects were happier to work for some salary than remaining unemployed for the same amount of money. It would be interesting to see whether such findings also hold in the developed world.

### 20.3.2. Data on product market slack in the United States

Data on product market slack are especially scarce in the United States. Given the importance of the issue, product market slack is not something that is well-measured or that the government tracks at all—probably because it is not part of standard macroeconomic theory. It would be valuable to collect more data on product market slack. It would be good to know how many goods remain unsold across the economy at a monthly frequency, and how many employees remain idle each month across the economy—waiting for customers to purchase their labor services.

It would also be good to know how many buyers there are for goods and services at any point in time, which would then allow us to measure product market tightness. Unlike on the labor market, where job vacancies are measured systematically, the desired amount of purchases on the product market is not recorded in any centralized dataset.

And while we have some idea of the cost of filling a job vacancy, there is very little data on the cost of visiting shops and filling help-wanted ads. Collecting some data on the matching cost on the product market would be very valuable—for instance, to calibrate the product market in the two-market business cycle model and to compute the efficient product market tightness.

### 20.3.3. Data on slack in other countries

The most important form of slack is unemployment, so the most important statistic is the number of unemployed workers at any point in time. Developed countries now collect such data, but many developing countries still do not. It seems critical to start measuring unemployment in these countries to track and stabilize business cycle fluctuations.

One challenge is that many workers are self-employed in the developing world, and workers who are self-employed are neither as idle as unemployed workers nor as occupied as formal workers, so they only partially contribute to slack in the economy. To complicate matters, Maloney (2004, p. 1161) reports that in Latin America about 30% of the self-employed are in the occupation involuntarily, while 70% choose to be self-employed voluntarily, enjoying the flexibility and independence from self-employment. Presumably, the involuntarily self-employed could be more productive and work longer hours if they were employed by firms, so they contribute more significantly to aggregate slack. The voluntarily self-employed may be producing and earning as much as they can in their occupation, and they might not contribute to aggregate slack at all. Keeping track of slack among the self-employed would probably require to measure unemployed hours. The unemployment rate among the self-employed would be the share of the time that they are idle.

To assess the tightness of foreign labor markets, it would be key to collect monthly data on job vacancies. To assess the full-employment rate of unemployment, as well as the unemployment gap, it would be important to measure the amount of labor devoted to recruiting in these labor markets. At the minimum, vacancies can be translated into recruiters by assuming some constant number of recruiters per vacancy; but it would be even better to have direct, monthly measures of the number of recruiters.

Of course, in addition to data on slack in the labor market, it would be extremely instructive to measure slack on the product market abroad. This would require measuring the slack rate in firms, small and large, as the Institute for Supply Management (ISM) does in the United States. The ISM asks firms how much more they could produce, given their current capital stock and workforce and organization, if only they had more demand for their product. In the slackish framework, this is the ideal information to compute the slack rate in the product market.

By processing the content of major newspapers—about 20,000 news articles per month—Caldara, Iacoviello, and Yu (2026, figure 2) construct an index of shortage in the United States from 1900 to 2024. The index includes shortages of food, of energy, in the industrial sector, and on the labor market. What is fascinating is that the episodes of labor market shortage that they measure perfectly overlap with the episodes during which labor market tightness is inefficiently high: World War 2, the Korean War, the Vietnam War, and the post-pandemic period. This suggests that it might be possible to obtain accurate measures of tightness in the labor market and possibly in the product market by textual analysis of newspapers: this might be a promising avenue to measure slack when government data are unavailable.

### 20.3.4. Estimating the rigidity of prices and wages

Finally, it would be valuable to use the large amount of price and wage data collected in the United States and abroad to estimate the rigidity of price and wage norms. This task would require redirecting current empirical efforts, which are guided by prevailing theories but do not estimate the statistics that are crucial to calibrate slackish models.

A large amount of price data are now available in many countries. However, most of the empirical effort is devoted to estimating "the average frequency and size of price changes" (Dhyne et al. 2006). In the New Keynesian paradigm, it is central to know how frequently prices adjust. In the slackish model, what is key is how strongly prices respond to shocks: what is the elasticity of prices with respect to underlying disturbances? that is, what is the passthrough of shocks into prices? More work is required to understand what the passthroughs are in different settings, and what influences them.

A similar comment applies to wages. A large amount of wage data have been collected in many countries as well. Most of the empirical effort is devoted to identifying downward nominal wage rigidity (Dickens et al. 2007). This is because Keynes (1936) postulated such rigidity was the main source of unemployment in recessions. But if firms know that they will not be able to cut wages, they will presumably be reluctant to raise them too. So logically, we would expect wages to be rigid in both directions, and in the slackish model such rigidity explains business cycle fluctuations. What is key is how strongly real wages respond to shocks: the elasticity of real wages to shocks. Few papers estimate this elasticity, so more work is required to understand how wages respond to shocks in different circumstances.

### 20.3.5. Estimating the slack dependence of policy multipliers

The slackish model is sharply state dependent. This implies that the effectiveness of policies varies sharply with the state of the economy. For example, in the slackish labor market model, we saw that when the labor market is slacker, public employment reduces unemployment more effectively, in-migration raises unemployment more sharply, and unemployment insurance—although it continues to reduce job-search effort—raises unemployment less. In the slackish business cycle model, when the economy is slacker, fiscal and monetary policy reduce unemployment more effectively.

It would be interesting to investigate these predictions systematically, in the United States and other countries. This investigation would help designing better policies—because it would help understanding when certain policies are particularly effective. The investigation would also be useful to evaluate the theory of slack.

Several papers offer promising starting points to estimate slack-dependent policy multipliers in the United States and abroad. For instance, Auerbach and Gorodnichenko (2013) estimate state-dependent fiscal multipliers in numerous OECD countries; Holden and Sparrman (2016) estimate state-dependent public-spending multipliers in 20 OECD countries; and Kopiec and Walerych (2026) estimate a slack-dependent monetary multiplier in Poland. These papers find that public spending and monetary policy reduce the unemployment rate more effectively when the economy is slack than when it is tight, in line with the predictions of the theory of slack.

## 20.4. Developing the theory of slack further

The theory of slack presented in the book captures essential mechanisms but remains deliberately stylized, leaving many natural extensions open for exploration. We discuss some extensions that might be especially fruitful here.

### 20.4.1. Microfounding the matching function

The central tool used to model slackish markets is a matching function. We saw that there exist microfoundations for standard matching functions. However, the microfoundations remain specific to the labor market. More work on the foundations of matching functions in generic markets would be especially valuable.

It would also be helpful to use this foundational work to understand better the shifts of the Beveridge curve on the labor market. The location of the Beveridge curve is determined by the efficacy of the labor market matching function. So understanding its microdeterminants would help understand the shifts of the Beveridge curve, and in turn the movements of the FERU. Typical shifts in the labor-market Beveridge curve do not affect the FERU much, but three large shifts over the past century had a sizable impact on the FERU—therefore on the unemployment target for the Federal Reserve. These shifts happened during dramatic periods of US history: the Great Depression, World War 2, and the coronavirus pandemic. It is unsurprising that such dramatic events shifted the curve, but it would be good to understand how that happened, and what the mechanism was.

### 20.4.2. Modeling price norms

In the models we have used fairly simple price and wage norms that were rigid. But we also saw that such rigidity is caused in large part by fairness concerns. Sellers do not want to charge prices that buyers find unfair, and firms do not want to pay wages that workers find unfair. It would be fruitful to incorporate more realistic price norms, built to mimic real-world fairness concerns, into the models. The concepts of fair wage developed by Akerlof (1980, 1982) and Akerlof and Yellen (1990), as well as the concept of fair price developed by Eyster, Madarasz, and Michaillat (2021), might be productive starting points.

Relatedly, it would be invaluable to have more estimates of the passthroughs of marginal valuation into prices, as well as estimates of the passthrough of marginal productivity into wages. Estimates of these passthroughs would help calibrating price and wage norms more realistically.

### 20.4.3. Studying out-of-equilibrium dynamics

When we considered dynamic markets, where trading relationships are formed and destroyed over time, we concentrated on the equilibrium—where market flows are balanced so slack is constant. It would be interesting to study out-of-equilibrium dynamics too. In the US labor market, flows are large, so out-of-equilibrium dynamics are fast and unlikely to matter much. But such dynamics may matter more on other US markets, or in other countries. For instance, labor market flows are much smaller in continental Europe (Elsby, Hobijn, and Sahin 2013, table 4). So there it would be important to study and incorporate out-of-equilibrium market dynamics.

### 20.4.4. Modeling international trade

An important aspect of the economy that is not discussed in the book but that could be included in future versions of the model is international trade. International trade fits well in the slackish framework: it takes time and effort for producers to find customers abroad, and it takes a lot of effort and money for buyers to find suppliers abroad. The geographical, cultural, and legal distance makes it difficult to trade with buyers or sellers in other countries—even more so in other continents (Egan and Mody 1992). It is because of these difficulties that sellers and buyers are keen to form long-lasting cross-country relationships (Egan and Mody 1992, table 3). The slackish framework might be helpful to insert international trade and global supply chains into a macroeconomic model, for instance following the approaches taken by Drozd and Nosal (2012) and Bai et al. (2026).

### 20.4.5. Modeling the housing market

Another market where slack is well documented is the housing market. There is always slack on the US housing market: hundreds of thousands of houses for sale but not sold in any month (Realtor.com 2025a). Another manifestation of the slackishness of the housing market is that it takes time to sell a house: the median time that a house stays on the market is always above 1 month, and sometimes as high as 3 months (Realtor.com 2025b). At the same time, we know that the housing market plays a key role in shaping business cycles (Feldstein 2007; Leamer 2007). Integrating the housing market into the slackish business cycle framework might provide interesting insights.

#### 20.4.6. Introducing capital

This book's focus has been on short-run issues, namely business cycles. We have not explored how economic slack might help us understand long-run issues, such as economic development and growth. Because of this short-run focus, we have concentrated on labor as a factor of production. We have not paid much attention to capital as a factor of production. We have barely looked at data on capital goods, and we have not explicitly modeled capital and related quantities, such as investment. Yet, capital also remains idle some of the time, and capital utilization varies over time (Gorodnichenko and Shapiro 2011). Moreover, slack in the form of low firm capacity utilization seems to be an important determinant of corporate investment (Grullon and Ikenberry 2025). The slackish framework might help connect capital utilization and capital accumulation.

Finally, capital goods are specialized, complex, and unique, so they take time and effort to sell and to buy.[2] As a result, physical capital faces a notable and countercyclical unemployment rate: Ottonello (2026) documents that the unemployment rate of capital goods in the United States averages 11% and is sharply countercyclical. The framework in the book might be helpful to understand trade in capital goods and slack in those markets.

#### 20.4.7. Introducing a credit market

Financial crises often include collapses of the credit market. A prime example is the Great Depression, when the stock market crash triggered a banking crisis and a generalized collapse of the credit market, which deepened and lengthened the depression (Bernanke 1983). To better understand cycles and especially their financial aspect, it would be useful to append a credit market to the business cycle model proposed here. Of course credit markets are slackish too and could be modeled as such: it is hard to find credit for borrowers, and hard for banks to find reputable borrowers, so that when they meet they stay in long term relationships (Wasmer and Weil 2004).

### 20.5. What causes business cycles?

Having seen how slack transmits aggregate demand shocks and how policy can mitigate them, the deeper question remains: what generates these demand shocks in the first

[2] See Gavazza (2011) for commercial airplanes and Ramey and Shapiro (2001) for aerospace equipment.

place? As my students always ask when I take questions at the end of the first lecture of my course on unemployment: “But why does unemployment always spike up after reaching rock bottom?” That’s a great question: why do we perennially have business cycles?

In this book we saw how unemployment would go up and down in response to shocks to aggregate demand, producing realistic business cycles. We also argued that stabilization policy might not have been conducted perfectly, because it might have ignored unemployment fluctuations or might have overestimated the FERU and so might have tolerated more unemployment than it should have. So monetary and fiscal policy could possibly have done better and smoothed out more forcefully the waves of unemployment.

But it still remains to explain what these fluctuations in aggregate demand are, where they come from, and why they seem to endlessly reappear to trigger the next recession. It seems that each recession has its own story: the 2020 recession was caused by the coronavirus pandemic; the Great Recession of 2007–2009 coincided with the global financial crisis; the 2001 recession followed the burst of the dot-com bubble; the 1990–1991 recession followed the Iraqi invasion of Kuwait; the twin 1980–1982 recessions were caused by the tight monetary policy imposed by Volcker; the Great Depression in 1929–1933 was triggered by the stock market crash of October 1929; and so on. In all these recessions the unemployment rate spiked. During the pandemic, it was probably unavoidable that a recession followed and unemployment spiked. But in all other cases, why did the events trigger a full-blown recession with a collapse in aggregate demand, why did similarly important global events not trigger a recession, and why are these downturns occurring like clockwork after periods of expansion and confidence?

There is no established answer to these questions, and much more work on it is warranted. One possibility is that some of the events unleash the “animal spirits” that lie dormant otherwise, as Keynes (1936, chapter 12) proposed:

> Even apart from the instability due to speculation, there is the instability due to the characteristic of human nature that a large proportion of our positive activities depend on spontaneous optimism rather than on a mathematical expectation, whether moral or hedonistic or economic. Most, probably, of our decisions to do something positive, the full consequences of which will be drawn out over many days to come, can only be taken as the result of animal spirits—a spontaneous urge to action rather than inaction, and not as the outcome of a weighted average of quantitative benefits multiplied by quantitative probabilities.

It might be that after periods of stability, the animal spirits always awaken and produce bubbles and then crashes. This could explain the dot-com recession and Great Depression (stock market bubbles), the Great Recession (housing market bubble), and maybe the next recession (bubble around artificial intelligence and cryptocurrency). Speculative manias, followed by financial crashes, have happened time and time again throughout history and around the world, driven by irrational exuberance from investors, and often exacerbated by fraudulent behavior.[3]

More generally, animal spirits might be an irrational fervor that drives economic decisions of crowds of people, as proposed by Akerlof and Shiller (2009). This fervor may follow certain events and not others, and when it is triggered, may lead to brash, large-scale reactions—consumers suddenly cutting spending and firms suddenly cutting hiring and traders suddenly dumping stocks. The slackish model provides a natural way to formalize how these swings in collective optimism translate into fluctuations in slack and output, and generate cycles.

It might even be that business cycles are not driven by exogenous shocks at all. Maybe business cycles are endogenous, not caused by shocks but due to circular or orbital dynamics around an equilibrium of the model. Diamond and Fudenberg (1989) find that such dynamics appear in a precursor of the slackish model developed by Diamond (1982), so endogenous cycles might appear in some versions of the slackish business cycle model, and may help explain recurring fluctuations in slack. To produce realistic endogenous cycles, it might be necessary to move away from the aggregate perspective taken in the book and port the slackish model to an agent-based framework.[4] This may be a more radical departure from conventional macroeconomics, but it should be doable given that the slackish model and agent-based models share similarities. An agent-based slackish model might help understand through which mechanism and under which conditions endogenous fluctuations might emerge—in the form of oscillatory or chaotic dynamics.[5]

[3] See for instance Akerlof and Romer (1993), Kindleberger and Alibert (2005), Shiller (2005), and Akerlof and Shiller (2015).

[4] See (Axtell and Farmer 2025) for a survey of the agent-based literature.

[5] Giardina and Bouchaud (2002) find such dynamics in an agent-based model of financial markets and Gualdi et al. (2015) in a macroeconomic agent-based model with unemployment.

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